Trigonometric Functions & the Unit Circle

Section 1: Radian Measure & Unit Circle Coordinates

1 What are the exact coordinates of the terminal point on the unit circle for angle $\theta = \frac{5\pi}{6}$?

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

2 Convert an angle of $\frac{7\pi}{4}$ radians into degree measure.

a. 315°

b. 225°

c. 300°

d. 135°

3 If the terminal ray of angle α passes through the point $(-3, -4)$, what is $\sin\alpha$?

a. $-\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{4}{3}$

d. $\frac{3}{5}$

Section 2: Exact Values of Trigonometric Functions

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4 Evaluate $\tan\left(\frac{4\pi}{3}\right)$.

a. $\sqrt{3}$

b. $-\sqrt{3}$

c. $\frac{\sqrt{3}}{3}$

d. $-\frac{\sqrt{3}}{3}$

5 What is the exact value of $\sec\left(-\frac{\pi}{4}\right)$?

a. $\sqrt{2}$

b. $-\sqrt{2}$

c. $\frac{\sqrt{2}}{2}$

d. 2

6 If $\sin\theta = \frac{3}{5}$ and $\cos\theta < 0$, find $\cot\theta$.

a. $-\frac{4}{3}$

b. $\frac{4}{3}$

c. $-\frac{3}{4}$

d. $-\frac{4}{5}$

Section 3: Sinusoidal Graphs & Transformations

7 Find the amplitude and period of the sinusoidal function $f(x) = -4\cos(3x - \pi) + 2$.

a. Amplitude $= 4$, Period $= \frac{2\pi}{3}$

b. Amplitude $= -4$, Period $= 3$

c. Amplitude $= 4$, Period $= 3\pi$

d. Amplitude $= 2$, Period $= \frac{2\pi}{3}$

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8 A Ferris wheel of radius 25 m completes one rotation every 60 seconds. Passengers board at the bottom, 2 m above the ground. Which cosine equation models height $h(t)$ in meters?

a. $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$

b. $h(t) = 25\cos\left(\frac{\pi}{30}t\right) + 27$

c. $h(t) = -25\cos\left(\frac{\pi}{60}t\right) + 2$

d. $h(t) = -27\cos\left(\frac{\pi}{30}t\right) + 25$

9 What is the phase shift of $g(x) = 2\sin\left(4x + \frac{\pi}{2}\right)$?

a. $\frac{\pi}{8}$ units to the left

b. $\frac{\pi}{8}$ units to the right

c. $\frac{\pi}{2}$ units to the left

d. $\frac{\pi}{4}$ units to the left

Section 4: Fundamental Trigonometric Identities

10 Simplify the trigonometric expression $\frac{\tan x}{\sec x}$.

a. $\sin x$

b. $\cos x$

c. $\csc x$

d. $\cot x$

11 Which expression is equivalent to $(1 - \cos^2\theta)(1 + \cot^2\theta)$ for all defined θ?

a. 1

b. $\sin^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

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12 Solve for $\theta \in [0, 2\pi)$: $2\cos^2\theta - \cos\theta - 1 = 0$.

a. $\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$

b. $\theta = \pi, \frac{\pi}{3}, \frac{5\pi}{3}$

c. $\theta = 0, \frac{\pi}{3}, \frac{5\pi}{3}$

d. $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

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